

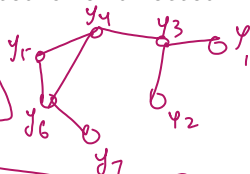
General framework for Parameter learning in graphical models

- Conditional distribution $\Pr(\mathbf{y}|\mathbf{x}, \theta)$, potentials are function of $\mathbf{x}$ and parameters θ to be learned.
- $\mathbf{y} = y_1, \dots, y_n$ forms a graphical model: ~~directed or undirected.~~
- Undirected:

$$\Pr(y_1, \dots, y_n | \mathbf{x}, \theta) = \frac{\prod_c \psi_c(\mathbf{y}_c, \mathbf{x}, \theta)}{Z_\theta(\mathbf{x})}$$

$$\psi_c(\mathbf{y}_c, \mathbf{x}, \theta) = e^{F_\theta(\mathbf{y}_c, \mathbf{x})}$$

$$= \frac{1}{Z_\theta(\mathbf{x})} \exp\left(\sum_c F_\theta(\mathbf{y}_c, \mathbf{x})\right)$$



where $Z_\theta(\mathbf{x}) = \sum_{\mathbf{y}} \exp(\sum_c F_\theta(\mathbf{y}_c, \mathbf{x}))$
 clique potential $\psi_c(\mathbf{y}_c, \mathbf{x}) = \exp(F_\theta(\mathbf{y}_c, \mathbf{x}))$

Computation challenge

Potentials coupled with a log-partition function or not.

- 1 No: **Closed form solutions**
- 2 Yes: Potentials attached to arbitrary overlapping subset of variables in a UDGM. Example = edge potentials in a grid graph. Solve using gradient descent kind of iterative algorithms.

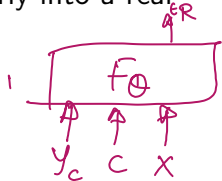
Forms of $F_{\theta}(\mathbf{y}_c, c, \mathbf{x})$

- Log-linear model over user-defined features. E.g. CRFs, Maxent models, etc.

Let K be number of features. Denote a feature as $f_k(\mathbf{y}_c, c, \mathbf{x})$.
Then,

$$F_{\theta}(\mathbf{y}_c, c, \mathbf{x}) = \sum_{k=1}^K \theta_k f_k(\mathbf{y}_c, c, \mathbf{x})$$

- Arbitrary function, e.g. a neural network that takes as input $\mathbf{y}_c, c, \mathbf{x}$ and transforms them possibly non-linearly into a real value. θ are the parameters of the network.

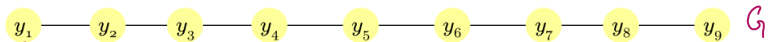


Example: Named Entity Recognition

→ My review of Fermat's last theorem by S. Singh

{Title, Author, other} $y_i \in$

t	1	2	3	4	5	6	7	8	9
x	My	review	of	Fermat's	last	theorem	by	S.	Singh
y	Other	Other	Other	Title	Title	Title	other	Author	Author



$$f(y_i, y_{i-1}, i, \mathbf{x})$$

$$Y_c = \{y_i, y_{i-1}\}$$

$$c = (i, i-1)$$

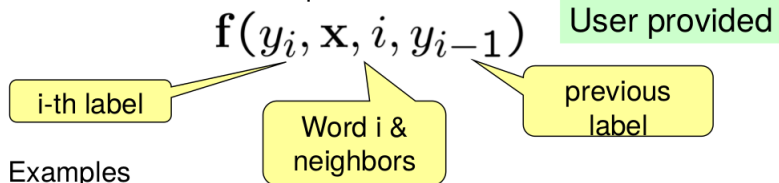
$$= (i)$$

Features decompose over adjacent labels.

$$f(\mathbf{x}, \mathbf{y}) = \sum_{i=1}^{|\mathbf{x}|} f(y_i, y_{i-1}, i, \mathbf{x})$$

Named Entity Recognition: Features

- Feature vector for each position



- Examples

$f_2(y_i, x, i, y_{i-1}) = \underline{1}$ if $\underline{y_i}$ is Person & $\underline{x_i}$ is Douglas

$f_3(y_i, x, i, y_{i-1}) = \underline{1}$ if $\underline{y_i}$ is Person & $\underline{y_{i-1}}$ is Other

Training

Given

- N input output pairs $D = \{(\underline{\mathbf{x}}^1, \underline{\mathbf{y}}^1), (\mathbf{x}^2, \mathbf{y}^2), \dots, (\mathbf{x}^N, \underline{\mathbf{y}}^N)\}$
- Form of F_θ
- Learn parameters θ by maximum likelihood.

$$\max_{\theta} \underline{LL}(\theta, D) = \max_{\theta} \sum_{i=1}^N \log \Pr(\underline{\mathbf{y}}^i | \underline{\mathbf{x}}^i, \theta)$$

Training undirected graphical model

$$\begin{aligned} LL(\theta, D) &= \sum_{i=1}^N \log \Pr(\mathbf{y}^i | \mathbf{x}^i, \theta) \\ &= \sum_{i=1}^N \log \frac{1}{Z_{\theta}(\mathbf{x}^i)} \exp\left(\sum_c F_{\theta}(\mathbf{y}_c^i, c, \mathbf{x}^i)\right) \\ &= \sum_i \left[\sum_c F_{\theta}(\mathbf{y}_c^i, c, \mathbf{x}^i) - \log Z_{\theta}(\mathbf{x}^i) \right] \end{aligned}$$

The first part is easy to compute but the second term requires to invoke an inference algorithm to compute $Z_{\theta}(\mathbf{x}^i)$ for each i . Computing the gradient of the above objective with respect to θ also requires inference.

Training via gradient descent

$$\mathbf{f} = [f_1 \ f_2 \ \dots \ f_n]$$
$$\theta_1 \ \theta_2 \ \dots \ \theta_n$$

Assume log-linear models like in CRFs where

$F_{\theta}(\mathbf{y}_c^i, c, \mathbf{x}^i) = \theta \cdot \mathbf{f}(\mathbf{x}^i, \mathbf{y}_c^i, c)$ Also, for brevity write

$$\mathbf{f}(\mathbf{x}^i, \mathbf{y}^i) = \sum_c \mathbf{f}(\mathbf{x}^i, \mathbf{y}_c^i, c)$$

Thus, we can write $\sum_c F_{\theta}(\mathbf{y}_c^i, c, \mathbf{x}^i) = \theta \mathbf{f}(\mathbf{x}^i, \mathbf{y}^i)$

where $\mathbf{f}$ is a vector of K features $[f_1, \dots, f_K]$

$$\log \Pr(\mathbf{y}^i | \mathbf{x}^i, \theta) = \theta \mathbf{f}(\mathbf{x}^i, \mathbf{y}^i) - \log Z_{\theta}(\mathbf{x}^i)$$

Training objective for log-linear models

$$\underline{LL(\theta)} = \sum_i \log \Pr(\underline{\mathbf{y}^i | \mathbf{x}^i}, \theta) = \sum_i (\underline{\theta \cdot \mathbf{f}(\mathbf{x}^i, \mathbf{y}^i)} - \log \underline{Z_\theta(\mathbf{x}^i)})$$

Add a regularizer to prevent over-fitting.

$$\underline{L(\theta)} = \max_{\theta} \sum_i (\underline{\theta \cdot \mathbf{f}(\mathbf{x}^i, \mathbf{y}^i)} - \log \underline{Z_\theta(\mathbf{x}^i)}) - \underline{\|\theta\|^2 / C}$$

Concave in θ $\implies$ gradient descent methods will work.

$$\nabla_{\theta} LL$$

$$\nabla_{\theta} \log Z_{\theta}(\mathbf{x}^i) = \log \sum_{\mathbf{y}} \frac{e^{\theta \cdot \mathbf{f}(\mathbf{x}^i, \mathbf{y})}}{Z_{\theta}(\mathbf{x}^i)}$$

Training log-linear models via gradient descent

$$\begin{aligned}\nabla L(\theta) &= \sum_i \mathbf{f}(\mathbf{x}^i, \mathbf{y}^i) - \frac{\sum_{\mathbf{y}'} \mathbf{f}(\mathbf{y}', \mathbf{x}^i) \exp(\theta \cdot \mathbf{f}(\mathbf{x}^i, \mathbf{y}'))}{Z_\theta(\mathbf{x}^i)} - 2\theta/C \\&= \sum_i \mathbf{f}(\mathbf{x}^i, \mathbf{y}^i) - \sum_{\mathbf{y}'} \mathbf{f}(\mathbf{x}^i, \mathbf{y}') \Pr(\mathbf{y}' | \theta, \mathbf{x}^i) - 2\theta/C \\&= \sum_i \mathbf{f}(\mathbf{x}^i, \mathbf{y}^i) - E_{\Pr(\mathbf{y}' | \theta, \mathbf{x}^i)} \mathbf{f}(\mathbf{x}^i, \mathbf{y}') - 2\theta/C\end{aligned}$$

Workout

Computing expected value of a feature in UGM

$$\begin{aligned}
 \mathbb{E}_{\Pr(\mathbf{y}'|\theta, \mathbf{x}^i)} f_k(\mathbf{x}^i, \mathbf{y}') &= \sum_{\mathbf{y}'} f_k(\mathbf{x}^i, \mathbf{y}') \Pr(\mathbf{y}'|\theta, \mathbf{x}^i) \\
 &= \sum_{\mathbf{y}'} \left(\sum_{\mathbf{c}} f_k(\mathbf{x}^i, \mathbf{y}'_{\mathbf{c}}, \mathbf{c}) \right) \Pr(\mathbf{y}'|\theta, \mathbf{x}^i) \\
 &= \sum_{\mathbf{c}} \sum_{\mathbf{y}'_{\mathbf{c}}} f_k(\mathbf{x}^i, \mathbf{y}'_{\mathbf{c}}, \mathbf{c}) \sum_{\mathbf{y}' - \mathbf{y}'_{\mathbf{c}}} \Pr(\mathbf{y}'|\theta, \mathbf{x}^i) \\
 &= \sum_{\mathbf{c}} \sum_{\mathbf{y}'_{\mathbf{c}}} f_k(\mathbf{x}^i, \mathbf{y}'_{\mathbf{c}}, \mathbf{c}) \Pr(\mathbf{y}'_{\mathbf{c}}|\theta, \mathbf{x}^i)
 \end{aligned}$$

$j=1$
 $\sum_{y_1, y_2, \dots, y_9} P(y_1, y_2, \dots, y_9 | x^i, \theta)$
 $= P(y_1, y_2 | x^i, \theta)$

$y_1, y_2, y_3, \dots, y_9$
 $\sum_{y_1, y_2, \dots, y_9} f_k(x^i, [y_1, \dots, y_9]) P(y_1, y_2, \dots, y_9 | x^i)$
 $\downarrow$
 $\sum_{y_1, \dots, y_9} \sum_{j=1}^8 f_k(x^i, [y_j, y_{j+1}], j) P(y_1, y_2, \dots, y_9 | x^i)$
 $\sum_{j=1}^8 \sum_{y_j, y_{j+1}} f_k(x^i, [y_j, y_{j+1}], j) \sum_{[y_1, \dots, y_9] - [y_j, y_{j+1}]} P(\mathbf{y}' | x^i, \theta)$

Computing $E_{\Pr(\mathbf{y}|\theta^t, \mathbf{x}^i)} f_k(\mathbf{x}^i, \mathbf{y})$

Three steps:

- 1 $\Pr(\mathbf{y}|\theta^t, \mathbf{x}^i)$ is represented as an undirected model where nodes are the different components of $\mathbf{y}$, that is $y_1, \dots, y_n$.

The potential $\psi_c(\mathbf{y}_c, \mathbf{x}, \theta)$ on clique c is $\exp(\theta^t \cdot \mathbf{f}(\mathbf{x}^i, \mathbf{y}_c, c))$

- 2 Run a sum-product inference algorithm on above UGM and compute for each $c, \mathbf{y}_c$ marginal probability $\mu(\mathbf{y}_c, c, \mathbf{x}^i)$.

- 3 Using these μ s we compute

$$E_{\Pr(\mathbf{y}|\theta^t, \mathbf{x}^i)} f_k(\mathbf{x}^i, \mathbf{y}) = \sum_c \sum_{\mathbf{y}_c} \mu(\mathbf{y}_c, c, \mathbf{x}^i) f_k(\mathbf{x}^i, c, \mathbf{y}_c)$$

Example: UGM with $K = 2$ features on three variables

$\mathbf{y} = [y_1 \ y_2 \ y_3]$ where each $y_j = +1$ or 0

Let the following two features be defined for it.



$f_1(\mathbf{x}, y_j, j) = x_j y_j$ (where x_j = intensity of pixel j)

$f_2(\mathbf{x}, (y_k, y_j), (k, j)) = 1$ if $y_k \neq y_j$ else 0

Initial parameters $\theta = [\theta_1, \theta_2] = [3, -2]$

Examples ($N = 1$): $\mathbf{x}^1 = [0.1, 0.7, 0.3]$, $\mathbf{y}^1 = [1, 1, 0]$

Using these we can calculate:

① $F_\theta(y_j, c = j, \mathbf{x}) = \text{Log Node potentials for } y_j = \theta \cdot f_1(\mathbf{x}, y_j, j) = \theta_1 x_j y_j$. For e.g. for y_1 it is $[0, 3 \times 0.1]$.

② $F_\theta((y_1, y_2), c = (1, 2), \mathbf{x}) = \log \text{ edge potentials} =$

$$\theta_2 f_2(\mathbf{x}, (y_1, y_2), (1, 2)) = \begin{matrix} y_2 & 0 & 1 \\ \begin{matrix} y_1=0 \\ 1 \end{matrix} & \begin{bmatrix} 0 & -2 \\ -2 & 0 \end{bmatrix} \end{matrix}$$

Example (continued)



- 1 Use above potentials to run sum-product inference on a junction tree to calculate marginals $\mu(y_j, j)$ and $\mu(y_k, y_j, (k, j))$

$$\mu(y_1, 1) \quad \mu(1, 1) \quad \mu(0, 2) \quad \mu_2(1, 2)$$

$$\mu(y_1, y_2, (0, 0)) \quad - \quad - \quad -$$

- 2 Using these we calculate expected value of features as:

$$E[f_1(\mathbf{x}^1, \mathbf{y})] = \sum_j x_j \mu_j(1, j) = \underline{0.1\mu(1, 1) + 0.7\mu(1, 2) + 0.3\mu(1, 3)}$$

$$\begin{aligned} E[f_2(\mathbf{x}^1, \mathbf{y})] &= \mu(1, 0, (1, 2)) + \mu(0, 1, (1, 2)) \\ &\quad + \mu(1, 0, (2, 3)) + \mu(0, 1, (2, 3)) \end{aligned}$$

Example (continued)

- ① The value of $\mathbf{f}(\mathbf{x}^1, \mathbf{y}^1)$ for each feature is (Note value of $\mathbf{y}^1 = [1, 1, 0]$):

$$f_1(\mathbf{x}^1, \mathbf{y}^1) = 0.1 * 1 + 0.7 * 1 + 0.3 * 0 = 0.8$$

$$f_2(\mathbf{x}^1, \mathbf{y}^1) = \mathbb{I}[y_1^1 \neq y_2^1] + \mathbb{I}[y_2^1 \neq y_3^1] = 1$$

- ② The gradient of each parameter is then.

$$\theta^t = [3, -2]$$

$$\nabla L(\theta_1) = 0.8 - E[f_1(\mathbf{x}^1, \mathbf{y})] - 2 * 3 / C$$

$$\nabla L(\theta_2) = 1 - E[f_2(\mathbf{x}^1, \mathbf{y})] + 2 * 2 / C$$

Another Example

Consider a parameter learning task for an undirected graphical model on six variables $\mathbf{y} = [y_1 \ y_2 \ y_3 \ y_4 \ y_5 \ y_6]$ where each $y_j = +1$ or -1 .

Let the following eight features be defined for it.

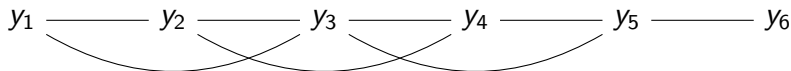
$$\begin{aligned} f_1(y_i, y_{i+1}) &= \mathbb{I}[y_i + y_{i+1} > 1], 1 \leq i < 5 & f_2(y_1, y_3) &= -2y_1y_3 \\ f_3(y_2, y_3) &= y_2y_3 & f_4(y_3, y_4) &= y_3y_4 \\ f_5(y_2, y_4) &= \mathbb{I}[y_2y_4 < 0] & f_6(y_4, y_5) &= 2y_4y_5 \\ f_7(y_3, y_5) &= -y_3y_5 & f_8(y_5, y_6) &= \mathbb{I}[y_5 + y_6 > 0]. \end{aligned}$$

where $\mathbb{I}[z] = 1$ if $z = \text{true}$ and 0 otherwise. That is,

$\mathbf{f}(\mathbf{y}) = [f_1 \ f_2 \ f_3 \ f_4 \ f_5 \ f_6 \ f_7 \ f_8]^T$. Assume the corresponding weight vector to be $\underline{\theta} = [1 \ 1 \ 1 \ 2 \ 2 \ 1 \ -1 \ 1]^T$

Example

Draw the underlying graphical model corresponding to the 6 variables.

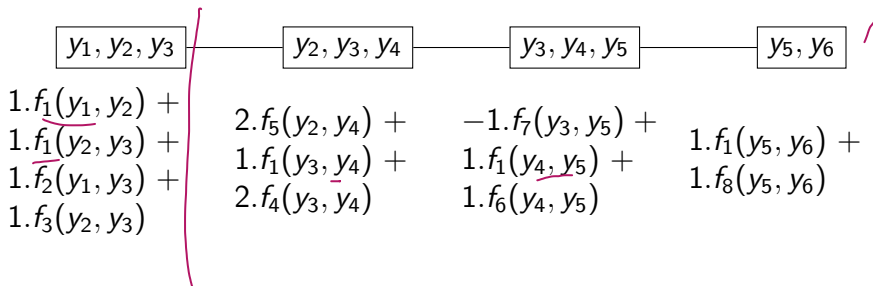


Draw an arc between any two y which appear together in any of the 8 features.

Example

Draw the junction tree corresponding to the graph above and assign potentials to each node of your junction tree so that you can run message passing on it to find $Z = \sum_{\mathbf{y}} \exp(\theta^T \mathbf{f}(\mathbf{x}, \mathbf{y}))$, that is, define $\psi_c(\mathbf{y}_c)$ in terms of the above quantities for each clique node c in the JT.

For clique c , $\psi_c(\mathbf{y}_c) = \exp(\theta \cdot \mathbf{f}_c(\mathbf{x}, \mathbf{y}_c))$. log of the potentials are shown below



Example

Suppose you use the junction tree above to compute the marginal probability for each pair of adjacent variables in the graph of part (a). Let $\mu_{ij}(-1, 1)$, $\mu_{ij}(1, 1)$, $\mu_{ij}(-1, -1)$, $\mu_{ij}(1, -1)$ denote the marginal probability of variable pairs y_i, y_j taking values $(-1, 1)$, $(1, 1)$, $(-1, -1)$ and $(1, -1)$ respectively. Express the expected value of the following features in terms of the μ values.

1

$$E[f_1] = \sum_j (f_1(-1, -1)\mu_{j,j+1}(-1, -1) + f_1(-1, 1)\mu_{j,j+1}(-1, 1) + f_1(1, -1)\mu_{j,j+1}(1, -1) + f_1(1, 1)\mu_{j,j+1}(1, 1))$$

$$2 \quad E[f_2] = 2(-\mu_{1,3}(-1, -1) + \mu_{1,3}(-1, 1) + \mu_{1,3}(1, -1) - \mu_{1,3}(1, 1))$$

$$3 \quad E[f_8] = \mu_{56}(1, 1)$$

Training algorithm

- 1: $\rightarrow$ Input: $D = \{(\mathbf{x}^i, \mathbf{y}^i)\}_{i=1}^N, \mathbf{f} : f_1 \dots f_K$
- 2: **Output:** $\theta = \operatorname{argmax} \sum_{i=1}^N (\theta \cdot \mathbf{f}(\mathbf{x}^i, \mathbf{y}^i) - \log Z_{\theta}(\mathbf{x}^i)) - \|\theta\|^2 / C$
- 3: Initialize $\theta^0 = \mathbf{0}$
- 4: **for** $t = 1 \dots T$ **do**
- 5: **for** $i = 1 \dots N$ **do**
- 6: $\rightarrow g_{k,i} = f_k(\mathbf{x}^i, \mathbf{y}^i) - E_{\Pr(\mathbf{y}'|\theta^t, \mathbf{x}^i)} f_k(\mathbf{x}^i, \mathbf{y}')$ $k = 1 \dots K$
- 7: **end for**
- 8: $g_k = \sum_i g_{k,i}$ $k = 1 \dots K$
- 9: $\rightarrow \theta_k^t = \theta_k^{t-1} + \gamma_t (g_k - 2\theta_k^{t-1} / C)$
- 10: **Exit if** $\|\mathbf{g}\| \approx \text{zero}$
- 11: **end for**

Running time of the algorithm is $O(INn(m^2 + K))$ where I is the total number of iterations.